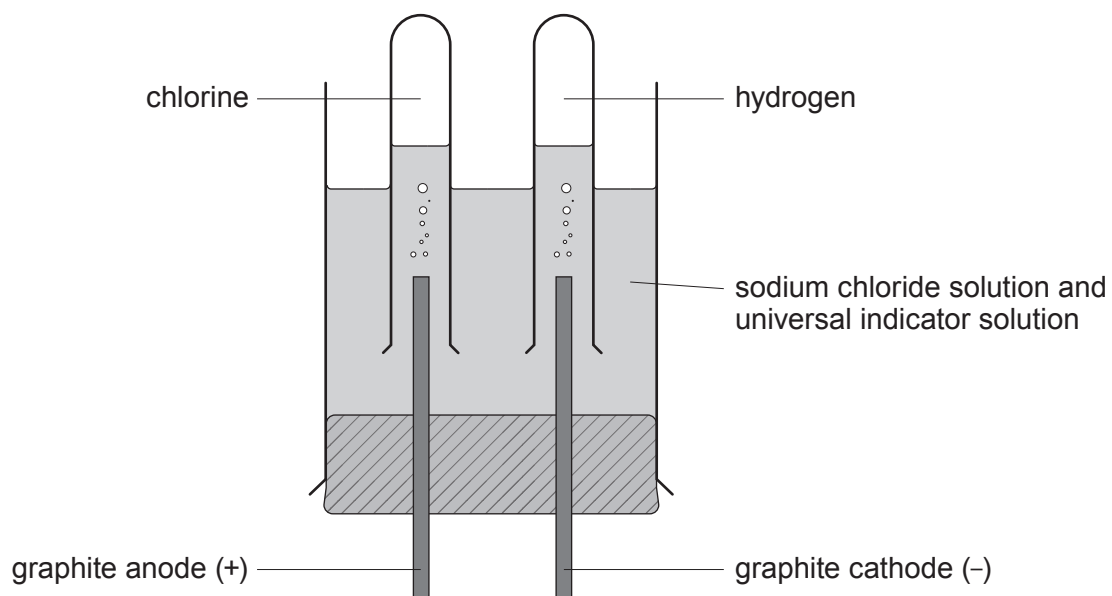


4. (a) The diagram shows apparatus that can be used for the electrolysis of sodium chloride solution.



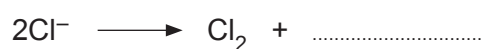
- (i) Explain why hydrogen, and **not** sodium, is formed at the cathode. [1]

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- (ii) Complete the electrode equation for the reaction at the anode. [1]



- (iii) Explain why the universal indicator turns purple during electrolysis. [2]

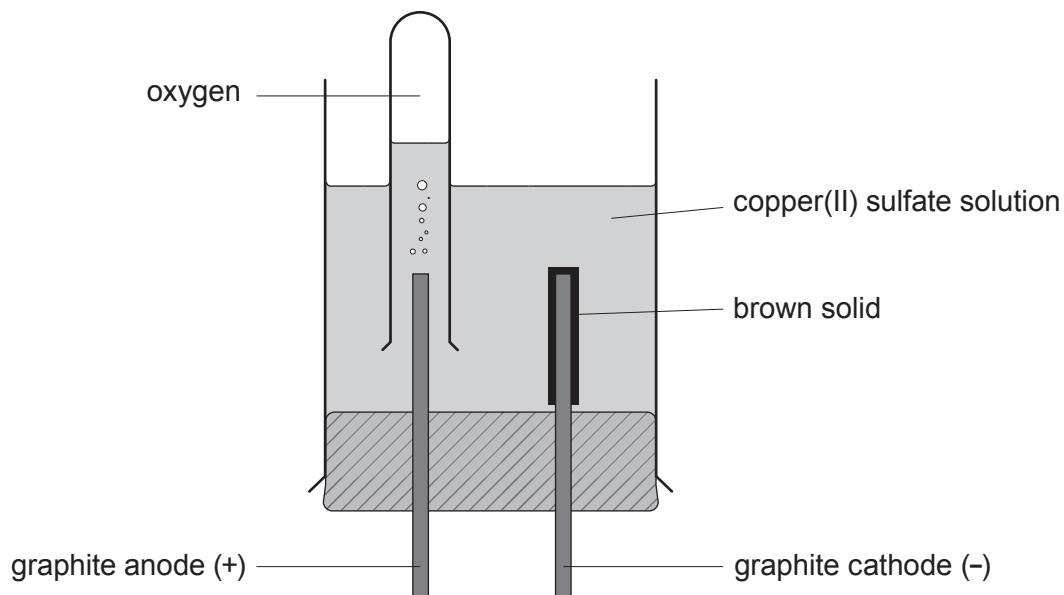
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(b) The diagram shows the electrolysis of copper(II) sulfate solution.



- (i) Explain, using the reaction occurring at the cathode, the meaning of the term *reduction*. [1]

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- (ii) Over time the electrolyte turns from blue to colourless. State the change you would make to the apparatus so that the electrolyte remains blue during the process. Give a reason for your answer. [2]

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- (iii) The electronic structure of oxygen is (2,6). Complete the diagram showing the outer shell electrons in an oxygen molecule, O_2 . [2]

